

KOMATSUJIMA AB, Japan

WMO: 478840

Lat: 34.005N

Lon: 134.626E

Elev: 6

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.0	-7.0	2.1	3.4	-6.0	2.3	4.1	14.4	7.2	13.4	7.1	6.1	300	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	32.9	26.0	31.9	25.6	30.9	25.5	26.7	30.8	26.3	30.2	25.9	29.6	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.4	20.6	29.1	25.1	20.2	28.8	24.8	19.9	28.5	83.9	31.1	81.9	30.5	80.1	29.7	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.4	11.9	10.7	-0.9	34.4	1.4	1.3	-2.0	35.3	-2.8	36.0	-3.6	36.7	-4.7	37.6	
			DB												
			WB	-3.1	27.7	0.9	0.7	-3.8	28.2	-4.3	28.6	-4.9	29.1	-5.5	29.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	6.6	7.0	9.6	14.6	19.2	22.9	26.7	28.0	25.0	19.8	14.2	9.1
	DBStd	7.82	2.12	2.51	2.76	2.91	2.07	2.05	1.97	1.49	2.46	2.44	2.83	2.63
	HDD10.0	290	107	91	41	2	0	0	0	0	0	0	2	48
	HDD18.3	1506	363	318	270	116	14	0	0	0	0	12	127	286
	CDD10.0	2830	2	6	29	140	287	388	518	559	449	304	128	21
	CDD18.3	1003	0	0	0	3	43	138	260	301	199	58	2	0
	CDH23.3	7940	0	0	0	6	88	583	2392	3280	1469	122	1	0
	CDH26.7	2356	0	0	0	0	6	78	739	1181	345	7	0	0
(14) Wind	WSAvg	5.4	6.3	6.2	6.1	5.6	5.1	4.3	4.4	4.6	5.0	5.2	5.5	6.1
(15) Precipitation	PrecAvg	1657	47	65	98	109	151	204	175	188	271	173	105	68
	PrecMax	2495	113	147	179	226	433	382	404	952	849	469	282	249
	PrecMin	992	7	14	46	41	45	55	63	22	68	19	15	6
	PrecStd	410	29	36	37	49	94	76	84	196	176	111	70	52
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	17.6	19.8	24.1	27.2	31.0	34.0	34.1	32.2	27.8	22.2	17.7
		MCWB	12.5	14.8	15.3	17.2	19.2	25.1	25.7	26.4	25.5	22.9	17.7	14.2
	2%	DB	12.3	14.3	17.9	22.1	25.2	28.8	32.2	33.0	31.0	25.9	21.0	15.5
		MCWB	8.9	11.1	14.2	17.0	18.8	23.5	25.6	26.2	25.1	21.4	17.2	11.8
	5%	DB	11.2	12.2	16.0	20.8	24.1	27.1	31.2	32.1	29.9	24.8	19.9	14.1
		MCWB	7.9	9.0	12.5	16.0	18.5	22.5	25.7	25.7	24.4	20.3	16.4	10.5
	10%	DB	10.1	11.0	14.7	19.2	23.0	26.1	30.2	31.2	28.8	23.5	18.8	13.1
		MCWB	6.7	7.9	11.3	15.0	18.1	22.2	25.4	25.6	23.9	19.2	15.4	9.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	15.2	16.6	19.9	22.7	25.8	27.1	27.6	26.7	23.9	19.7	15.4
		MCDB	14.8	17.1	18.3	22.1	24.6	29.5	31.3	32.0	30.1	26.5	21.0	16.7
	2%	WB	10.3	11.8	14.9	18.3	21.1	24.7	26.5	26.9	26.0	22.3	18.2	12.7
		MCDB	11.6	13.8	17.1	20.7	23.0	27.3	30.5	31.2	29.3	24.8	20.1	14.9
	5%	WB	8.5	10.0	13.2	16.9	19.9	24.1	26.1	26.6	25.4	21.1	16.8	11.2
		MCDB	10.7	11.6	15.5	19.6	22.3	26.2	29.9	30.6	28.5	23.4	19.2	13.4
	10%	WB	7.0	8.5	11.6	15.7	19.0	23.3	25.7	26.2	24.7	19.9	15.6	10.1
		MCDB	9.6	10.5	14.0	18.5	22.1	25.4	29.3	30.0	27.7	22.8	18.2	12.5
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.9	6.8	7.1	6.8	5.7	5.7	6.2	5.4	5.3	5.6	5.4
		MCDBR	7.2	7.9	9.1	9.1	8.7	7.0	7.3	7.4	6.6	6.3	6.7	6.8
		MCWBR	5.5	6.2	7.1	5.6	4.7	3.4	3.0	3.3	3.2	3.8	4.7	5.0
	5% WB	MCDBR	6.9	7.3	8.4	7.5	6.9	5.6	6.5	6.7	5.7	5.3	6.3	6.5
		MCWBR	6.2	6.6	7.6	5.6	4.7	3.2	3.1	3.3	3.1	3.9	5.3	5.5
(40) Clear-Sky Solar Irradiance	taub		0.399	0.447	0.508	0.540	0.547	0.562	0.561	0.531	0.496	0.459	0.423	0.393
	taud		2.175	2.056	1.911	1.883	1.938	1.976	2.025	2.098	2.167	2.187	2.193	2.208
	Ebn at Noon		783	783	766	764	763	750	747	763	771	765	752	764
	Edh at Noon		117	147	184	198	189	182	172	158	141	128	115	108
(44) All-Sky Solar Radiation	RadAvg		2.58	3.17	4.13	4.87	5.19	4.58	5.13	5.31	4.12	3.36	2.63	2.32
	RadStd		0.18	0.24	0.29	0.48	0.52	0.48	0.61	0.47	0.35	0.35	0.23	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page